

Qualification Pack

Embedded System Developer (A-Level Embedded System Design)

QP Code: NIE/ELE/Q0202

Version: 1.0

NSQF Level: 5

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:h.sihare@nielit.gov.in

Qualification Pack

Contents

NIE/ELE/Q0202: Embedded System Developer (A-Level Embedded System Design)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	4
NIE/ELE/N0206: Embedded System Design & C- Programming	6
NIE/ELE/N0207: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming	10
NIE/ELE/N0208: Sensor data acquisition and actuation in embedded systems	15
NIE/ELE/N0209: Linux Internals	19
NIE/ELE/N0210: RTOS for Embedded Application	23
NIE/ELE/N0211: Embedded Communication Protocols (UART, SPI, I2C, CAN, USB)	28
NIE/ELE/N0212: Wireless Sensor networks	32
NIE/ELE/N0213: Linux Device Driver Development	37
NIE/ELE/N0214: Internet of Things	42
NIE/ELE/N0215: AI for Embedded Application	46
NIE/ELE/N0219: Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver	50
DGT/VSQ/N0103: Employability Skills (90 Hours)	54
Assessment Guidelines and Weightage	62
<i>Assessment Guidelines</i>	62
<i>Assessment Weightage</i>	63
Acronyms	65
Glossary	66

Qualification Pack

NIE/ELE/Q0202: Embedded System Developer (A-Level Embedded System Design)

Brief Job Description

This course is a comprehensive training program designed to build proficiency in embedded systems, focusing on hardware and software components. It covers topics such as Embedded C programming, ARM Cortex-M microcontroller architecture, and Linux internals. Participants will also explore real-time operating systems (RTOS), embedded communication protocols, and wireless sensor networks. The course emphasizes hands-on learning with practical applications in areas like IoT and AI for embedded systems, making learners industry-ready in embedded technology.

Personal Attributes

An Embedded System Developer requires strong problem-solving skills to design and troubleshoot embedded systems efficiently. They must pay attention to detail to ensure accurate coding and hardware integration. Adaptability is key, as they need to stay updated with evolving technologies and tools. Effective communication and teamwork are essential to collaborate with cross-functional teams and convey technical concepts clearly.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0206: Embedded System Design & C- Programming](#)
2. [NIE/ELE/N0207: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming](#)
3. [NIE/ELE/N0208: Sensor data acquisition and actuation in embedded systems](#)
4. [NIE/ELE/N0209: Linux Internals](#)
5. [NIE/ELE/N0210: RTOS for Embedded Application](#)
6. [NIE/ELE/N0211: Embedded Communication Protocols \(UART, SPI, I2C, CAN, USB\)](#)
7. [NIE/ELE/N0212: Wireless Sensor networks](#)
8. [NIE/ELE/N0213: Linux Device Driver Development](#)
9. [NIE/ELE/N0214: Internet of Things](#)
10. [NIE/ELE/N0215: AI for Embedded Application](#)
11. [NIE/ELE/N0219: Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols \(UART, SPI, I2C, CAN, USB\), Wireless Sensor Networks, and Device Driver](#)

Qualification Pack

12. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	Embedded System Design
Country	India
NSQF Level	5
Credits	26
Aligned to NCO/ISCO/ISIC Code	5
Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR UG in relevant field (Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Completed 2nd year diploma after 12th (Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA

Qualification Pack

Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-05-EH-02592-2024-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ELE/N0206: Embedded System Design & C- Programming

Description

This NOS equips the students to explore the characteristics and components of embedded systems, setting up the development environment, and mastering C language syntax and basic programming constructs

Scope

The scope covers the following :

- The scope of this course encompasses an in-depth study of embedded systems and their characteristics, focusing on the C programming language tailored for embedded applications. Students will learn to set up the development environment and master C language syntax, variables, data types, control flow, pointers, and memory management techniques. The course also covers input/output operations, bit-level manipulation, peripheral register control, interrupt handling, and timer configurations. This comprehensive approach ensures students gain practical skills and theoretical knowledge essential for embedded systems programming.

Elements and Performance Criteria

Embedded Systems and C Programming

To be competent, the user/individual on the job must be able to:

- PC1.** Comprehend the fundamental characteristics and applications of embedded systems
- PC2.** Gain proficiency in the basics of the C programming language tailored for embedded systems
- PC3.** Successfully set up and configure an embedded systems development environment, including necessary tools and software

Programming Constructs and Memory Management

To be competent, the user/individual on the job must be able to:

- PC4.** Demonstrate understanding of C language syntax, including variables, data types, operators, and control flow structures
- PC5.** Apply pointers in various programming scenarios, understanding their significance in embedded systems
- PC6.** Efficiently use dynamic memory allocation and deallocation techniques, and implement memory management strategies for optimal performance in embedded systems

Peripheral Control and Interrupt Handling

To be competent, the user/individual on the job must be able to:

- PC7.** Access and manipulate peripheral registers to control hardware components
- PC8.** Implement interrupt handling mechanisms and develop interrupt service routines for responsive embedded applications
- PC9.** Configure timers and manage interrupt-based timer operations to achieve precise timing and scheduling in embedded systems

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Know the steps and tools required to set up a development environment for embedded systems programming, including IDEs, compilers, and debuggers.
- KU2.** Gain in-depth knowledge of variables, data types, operators, control flow structures (if, switch, loops), and their application in embedded systems programming
- KU3.** Learn the configurations and operations of timers in embedded systems, including how to manage interrupts and implement interrupt-based timing mechanisms

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to identify, analyze, and solve problems in embedded systems by using debugging tools and techniques effectively.
- GS2.** Enhance communication and teamwork skills by collaborating with peers on projects, sharing knowledge, and providing constructive feedback
- GS3.** Foster adaptability by staying current with new tools, technologies, and methodologies in the rapidly evolving field of embedded systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Embedded Systems and C Programming</i>	30	25	-	6
PC1. Comprehend the fundamental characteristics and applications of embedded systems	-	-	-	-
PC2. Gain proficiency in the basics of the C programming language tailored for embedded systems	-	-	-	-
PC3. Successfully set up and configure an embedded systems development environment, including necessary tools and software	-	-	-	-
<i>Programming Constructs and Memory Management</i>	30	25	-	7
PC4. Demonstrate understanding of C language syntax, including variables, data types, operators, and control flow structures	-	-	-	-
PC5. Apply pointers in various programming scenarios, understanding their significance in embedded systems	-	-	-	-
PC6. Efficiently use dynamic memory allocation and deallocation techniques, and implement memory management strategies for optimal performance in embedded systems	-	-	-	-
<i>Peripheral Control and Interrupt Handling</i>	40	30	-	7
PC7. Access and manipulate peripheral registers to control hardware components	-	-	-	-
PC8. Implement interrupt handling mechanisms and develop interrupt service routines for responsive embedded applications	-	-	-	-
PC9. Configure timers and manage interrupt-based timer operations to achieve precise timing and scheduling in embedded systems	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0206
NOS Name	Embedded System Design & C- Programming
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0207: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming

Description

This NOS equips the students to have the skills necessary to develop, optimize, and manage embedded applications, ensuring efficient and effective use of microcontroller resources.

Scope

The scope covers the following :

- This NOS provides a comprehensive understanding of Embedded C programming, embedded systems concepts, and ARM Cortex microcontroller architectures. Students will learn the fundamentals and advanced concepts of C programming, including memory management, pointers, and file operations. They will gain insights into the application areas, architecture, and development tools for embedded systems, with a focus on recent trends. The course covers ARM Cortex architectures in depth, including the M4 and A processors, and their development environments.

Elements and Performance Criteria

Embedded C Programming

To be competent, the user/individual on the job must be able to:

- PC1.** ● Demonstrate proficiency in writing and understanding complex C programs involving control flow, data structures, and memory management.
- PC2.** ● Implement algorithms using pointers and understand their role in memory manipulation and efficient programming.
- PC3.** ● Develop error-free programs with correct usage of bitwise operators and typecasting for optimized code execution

ARM Cortex Architecture

To be competent, the user/individual on the job must be able to:

- PC4.** ● Demonstrate the ability to configure and utilize Cortex M4 peripherals effectively in embedded applications, including GPIO, timers, UARTs, and ADCs
- PC5.** ● Develop and debug assembly language programs for ARM Cortex-M4 processors, ensuring proper memory access and interrupt handling
- PC6.** ● Design and implement interrupt service routines (ISRs) for real-time event management on Cortex M4 microcontrollers

Power Management in Embedded Systems

To be competent, the user/individual on the job must be able to:

- PC7.** ● Design power optimization strategies using various power modes like Active, Sleep, and Standby to extend battery life and improve energy efficiency
- PC8.** ● Implement clock management techniques, including clock gating and dynamic frequency scaling, to minimize power consumption during idle periods
- PC9.** ● Develop energy profiling techniques and utilize power management APIs to optimize energy usage and enhance system performance

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Gain knowledge of advanced topics such as conditional compilation, pre-processor directives, and file operations in C programming
- KU2.** Comprehend the architecture and organization of ARM Cortex processors, including the Cortex-M4 register set, instruction set architecture (ISA), and memory hierarchy
- KU3.** Understand the principles of power management in embedded systems, including power modes and clock management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to analyze complex problems in embedded systems programming and find efficient solution using C programming, ARM Cortex architecture, and power management strategies
- GS2.** Enhance critical thinking skills by evaluating different approaches to memory management, peripheral control, and power optimization in embedded systems development
- GS3.** Improve communication skills by documenting code, explaining programming concepts, and presenting project findings effectively to peers and stakeholders in the field of embedded systems and microcontroller programming

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Embedded C Programming</i>	30	25	-	6
PC1. ● Demonstrate proficiency in writing and understanding complex C programs involving control flow, data structures, and memory management.	-	-	-	-
PC2. ● Implement algorithms using pointers and understand their role in memory manipulation and efficient programming.	-	-	-	-
PC3. ● Develop error-free programs with correct usage of bitwise operators and typecasting for optimized code execution	-	-	-	-
<i>ARM Cortex Architecture</i>	30	25	-	7
PC4. ● Demonstrate the ability to configure and utilize Cortex M4 peripherals effectively in embedded applications, including GPIO, timers, UARTs, and ADCs	-	-	-	-
PC5. ● Develop and debug assembly language programs for ARM Cortex-M4 processors, ensuring proper memory access and interrupt handling	-	-	-	-
PC6. ● Design and implement interrupt service routines (ISRs) for real-time event management on Cortex M4 microcontrollers	-	-	-	-
<i>Power Management in Embedded Systems</i>	40	30	-	7
PC7. ● Design power optimization strategies using various power modes like Active, Sleep, and Standby to extend battery life and improve energy efficiency	-	-	-	-
PC8. ● Implement clock management techniques, including clock gating and dynamic frequency scaling, to minimize power consumption during idle periods	-	-	-	-
PC9. ● Develop energy profiling techniques and utilize power management APIs to optimize energy usage and enhance system performance	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0207
NOS Name	Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0208: Sensor data acquisition and actuation in embedded systems

Description

The objective is to understand transducers' role, sensor classifications, principles of transduction, and sensor characteristics

Scope

The scope covers the following :

- This course offers an in-depth introduction to transducers, sensors, and actuators, focusing on their roles in measurement and control systems. Students will learn about the principles of transduction, sensor and actuator classifications, characteristics, and applications in various fields. The course also covers interfacing these components with microcontrollers, signal conditioning, and calibration techniques. By the end, students will be equipped to implement sensors and actuators in real-world applications, enhancing their skills in modern measurement and control technologies

Elements and Performance Criteria

Sensor Fundamentals and Types

To be competent, the user/individual on the job must be able to:

- PC1.** ● Ability to explain the basic principles of sensors and their functions accurately
- PC2.** ● Demonstrate proficiency in classifying sensors based on measured quantities and sensing mechanisms
- PC3.** ● Evaluate sensor characteristics such as sensitivity, accuracy, linearity, and range in practical scenarios

Actuator Fundamentals and Applications

To be competent, the user/individual on the job must be able to:

- PC4.** ● Demonstrate understanding of actuator functions and classify them based on output types and characteristics
- PC5.** ● Apply knowledge of actuators to solve real-world problems in robotics, automation, and fluid control
- PC6.** ● Evaluate and select appropriate actuators for specific applications based on performance criteria like response time, force, torque, and displacement

Integration and Interfacing Techniques

To be competent, the user/individual on the job must be able to:

- PC7.** ● Successfully interface sensors and actuators with microcontrollers and embedded systems
- PC8.** ● Apply signal conditioning and amplification techniques effectively to improve sensor and actuator performance
- PC9.** ● Demonstrate calibration and compensation methods to ensure accurate integration and reliable functioning in control systems

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understand the role of transducers in measurement and control systems
- KU2.** Understanding Actuators and their Applications-Comprehend the functions and classifications of actuators according to their output types and characteristics
- KU3.** Understand the integration of sensors and actuators with microcontrollers and embedded systems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze sensor and actuator requirements in different applications to propose suitable solutions.
- GS2.** Conduct experiments to validate sensor accuracy, linearity, and range under different conditions
- GS3.** Effectively communicate technical concepts related to transducers, sensors, and actuators

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sensor Fundamentals and Types</i>	30	25	-	6
PC1. ● Ability to explain the basic principles of sensors and their functions accurately	-	-	-	-
PC2. ● Demonstrate proficiency in classifying sensors based on measured quantities and sensing mechanisms	-	-	-	-
PC3. ● Evaluate sensor characteristics such as sensitivity, accuracy, linearity, and range in practical scenarios	-	-	-	-
<i>Actuator Fundamentals and Applications</i>	30	25	-	7
PC4. ● Demonstrate understanding of actuator functions and classify them based on output types and characteristics	-	-	-	-
PC5. ● Apply knowledge of actuators to solve real-world problems in robotics, automation, and fluid control	-	-	-	-
PC6. ● Evaluate and select appropriate actuators for specific applications based on performance criteria like response time, force, torque, and displacement	-	-	-	-
<i>Integration and Interfacing Techniques</i>	40	30	-	7
PC7. ● Successfully interface sensors and actuators with microcontrollers and embedded systems	-	-	-	-
PC8. ● Apply signal conditioning and amplification techniques effectively to improve sensor and actuator performance	-	-	-	-
PC9. ● Demonstrate calibration and compensation methods to ensure accurate integration and reliable functioning in control systems	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0208
NOS Name	Sensor data acquisition and actuation in embedded systems
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0209: Linux Internals

Description

This Course is to provide a comprehensive understanding of Embedded Linux, focusing on its architecture, kernel basics, system calls, and file and process management. Students will learn about memory and device management, inter-process communication, multi-threading, synchronization, and networking concepts specific to embedded systems.

Scope

The scope covers the following :

- This course encompasses the fundamental principles and practical applications of Embedded Linux. It aims to provide students with a thorough understanding of how Linux is integrated into embedded systems. The course covers key aspects of Linux kernel architecture, system calls, file and process management, memory and device management, inter-process communication, multi-threading, synchronization, and networking in embedded environments. By the end of the course, students will be proficient in the concepts and practices necessary for developing and managing embedded systems using Linux.

Elements and Performance Criteria

Embedded Linux Architecture and Components

To be competent, the user/individual on the job must be able to:

- PC1.** ● Students should be able to explain the overall structure of Embedded Linux and identify its key components
- PC2.** ● Students should accurately describe the differences between Embedded Linux and standard Linux distributions
- PC3.** ● Analyze and discuss case studies where Embedded Linux is used, highlighting its advantages and challenges

Linux Kernel Basics

To be competent, the user/individual on the job must be able to:

- PC4.** ● Students should demonstrate how the kernel manages resources such as processes, memory, and devices
- PC5.** ● Perform practical exercises that involve configuring and optimizing kernel parameters for resource management
- PC6.** ● Write simple programs that make use of system calls to perform basic tasks like file operations or process management

Inter-process Communication and Multi-threading

To be competent, the user/individual on the job must be able to:

- PC7.** ● Write and test programs that utilize IPC to communicate between processes
- PC8.** ● Develop and debug multi-threaded applications, ensuring proper synchronization and resource sharing

Qualification Pack

- PC9.** ● Analyze and present case studies that involve complex IPC and multi-threading scenarios, discussing the solutions and their impact on system performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the hierarchical structure of Embedded Linux, including the kernel, libraries, and application layers
- KU2.** Familiarity with the roles and functions of each component within the Embedded Linux architecture
- KU3.** Comprehensive understanding of the Linux kernel architecture, including its layers and modules
- KU4.** Knowledge of how different parts of the kernel interact and function together
- KU5.** Comprehensive knowledge of multi-threading concepts and techniques in Linux
- KU6.** Skills to develop, debug, and optimize multi-threaded applications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Skills in presenting technical concepts and solutions to diverse audiences, including non-technical stakeholders
- GS2.** Strong ability to identify, diagnose, and resolve issues within embedded Linux systems.
- GS3.** Proficiency in using debugging tools and techniques to fix kernel, application, and system-level problems.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Embedded Linux Architecture and Components</i>	30	25	-	6
PC1. ● Students should be able to explain the overall structure of Embedded Linux and identify its key components	-	-	-	-
PC2. ● Students should accurately describe the differences between Embedded Linux and standard Linux distributions	-	-	-	-
PC3. ● Analyze and discuss case studies where Embedded Linux is used, highlighting its advantages and challenges	-	-	-	-
<i>Linux Kernel Basics</i>	30	25	-	7
PC4. ● Students should demonstrate how the kernel manages resources such as processes, memory, and devices	-	-	-	-
PC5. ● Perform practical exercises that involve configuring and optimizing kernel parameters for resource management	-	-	-	-
PC6. ● Write simple programs that make use of system calls to perform basic tasks like file operations or process management	-	-	-	-
<i>Inter-process Communication and Multi-threading</i>	40	30	-	7
PC7. ● Write and test programs that utilize IPC to communicate between processes	-	-	-	-
PC8. ● Develop and debug multi-threaded applications, ensuring proper synchronization and resource sharing	-	-	-	-
PC9. ● Analyze and present case studies that involve complex IPC and multi-threading scenarios, discussing the solutions and their impact on system performance	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0209
NOS Name	Linux Internals
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0210: RTOS for Embedded Application

Description

The course delves into Real-Time Systems, covering key concepts such as hard and soft real-time requirements, RTOS characteristics like task scheduling and interrupt handling, and memory management in RTOS environments. It also explores power management strategies for embedded systems and real-time system design considerations through hands-on projects, fostering a comprehensive understanding of real-time embedded systems development.

Scope

The scope covers the following :

- This course encompasses an introduction to real-time systems and their distinctive characteristics like hard real-time and soft real-time requirements. It delves into real-time constraints, deadlines, and response times, exploring Real-Time Operating Systems (RTOS) and their pivotal role in embedded applications. Students will gain insights into task scheduling and management, interrupt handling, device drivers, inter-task communication, memory management, and power management strategies specific to real-time embedded systems. The course also includes practical exercises and projects focusing on RTOS-based applications to reinforce learning and application skills.

Elements and Performance Criteria

Real-Time Systems Overview

To be competent, the user/individual on the job must be able to:

- PC1.** ● Ability to differentiate between hard and soft real-time requirements and understand their implications on system design
- PC2.** ● Demonstrate knowledge of real-time constraints, deadlines, and response times in various application scenarios
- PC3.** ● Analyze and assess real-world examples to determine the appropriate real-time system characteristics required for different applications

Real-Time Operating Systems (RTOS)

To be competent, the user/individual on the job must be able to:

- PC4.** ● Successfully implement and evaluate task scheduling algorithms such as priority-based, round-robin, and rate-monotonic scheduling
- PC5.** ● Demonstrate proficiency in handling interrupts, managing system resources effectively, and optimizing memory allocation in an RTOS environment
- PC6.** ● Design and develop real-time applications using RTOS principles, including task creation, management, synchronization, and resource utilization

Real-Time System Design

To be competent, the user/individual on the job must be able to:

- PC7.** ● Evaluate and compare power management strategies and techniques for embedded systems, considering trade-offs between performance and energy efficiency
- PC8.** ● Apply context switching and task synchronization techniques in practical scenarios, ensuring system stability and real-time responsiveness

Qualification Pack

- PC9.** ● Successfully complete hands-on exercises and projects that involve designing and implementing real-time systems using RTOS, demonstrating effective problem-solving and practical application of theoretical knowledge

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of real-time system characteristics, including hard and soft real-time requirements, deadlines, and response times.
- KU2.** Familiarity with memory management techniques, inter-task communication, synchronization methods, and power management strategies specific to RTOS environments
- KU3.** Ability to analyze real-time embedded systems requirements, design efficient real-time applications, and implement hands-on projects using RTOS principles

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze and solve complex problems related to real-time system design, task scheduling, and resource management
- GS2.** Effective communication of real-time system design considerations, task scheduling strategies, and memory management techniques
- GS3.** Proficiency in using real-time operating systems, including task creation, scheduling, and synchronization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Real-Time Systems Overview</i>	30	25	-	6
PC1. ● Ability to differentiate between hard and soft real-time requirements and understand their implications on system design	-	-	-	-
PC2. ● Demonstrate knowledge of real-time constraints, deadlines, and response times in various application scenarios	-	-	-	-
PC3. ● Analyze and assess real-world examples to determine the appropriate real-time system characteristics required for different applications	-	-	-	-
<i>Real-Time Operating Systems (RTOS)</i>	30	25	-	7
PC4. ● Successfully implement and evaluate task scheduling algorithms such as priority-based, round-robin, and rate-monotonic scheduling	-	-	-	-
PC5. ● Demonstrate proficiency in handling interrupts, managing system resources effectively, and optimizing memory allocation in an RTOS environment	-	-	-	-
PC6. ● Design and develop real-time applications using RTOS principles, including task creation, management, synchronization, and resource utilization	-	-	-	-
<i>Real-Time System Design</i>	40	30	-	7
PC7. ● Evaluate and compare power management strategies and techniques for embedded systems, considering trade-offs between performance and energy efficiency	-	-	-	-
PC8. ● Apply context switching and task synchronization techniques in practical scenarios, ensuring system stability and real-time responsiveness	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. ● Successfully complete hands-on exercises and projects that involve designing and implementing real-time systems using RTOS, demonstrating effective problem-solving and practical application of theoretical knowledge	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0210
NOS Name	RTOS for Embedded Application
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0211: Embedded Communication Protocols (UART, SPI, I2C, CAN, USB)

Description

This NOS provides an in-depth exploration of embedded systems and communication protocols, covering fundamentals and hands-on implementation of UART, SPI, I2C, CAN, USB, and Ethernet communication. Students will learn about the principles, configurations, and practical interfacing of these communication methods, enhancing their skills in developing robust and efficient embedded systems. Emphasis is placed on understanding data communication basics and integrating various communication standards in real-world applications.

Scope

The scope covers the following :

- The scope of this course encompasses a comprehensive study of embedded systems and their communication protocols. It covers the fundamentals of data communication, including serial and parallel communication techniques. Students will delve into UART, SPI, I2C, CAN, USB, and Ethernet communication standards, understanding their principles, configurations, and practical implementations. The course includes hands-on programming and interfacing exercises, preparing students to design and develop embedded systems with various communication capabilities for real-world applications.

Elements and Performance Criteria

UART (Universal Asynchronous Receiver-Transmitter) Communication

To be competent, the user/individual on the job must be able to:

- PC1.** • Demonstrate understanding of UART communication principles, including start/stop bits and data frames
- PC2.** • Utilize software tools to configure UART settings on various microcontrollers
- PC3.** • Write and debug code to implement UART communication between a microcontroller and peripheral devices

SPI (Serial Peripheral Interface) Communication

To be competent, the user/individual on the job must be able to:

- PC4.** • Explain the principles and components of SPI communication, including master/slave roles and data exchange mechanisms
- PC5.** • Write and test code to enable SPI communication between microcontrollers and peripherals
- PC6.** • Develop a project showcasing effective SPI communication in an embedded system application

I2C (Inter-Integrated Circuit) Communication

To be competent, the user/individual on the job must be able to:

- PC7.** ● Describe the fundamentals of I2C communication, including addressing and data transfer mechanisms
- PC8.** ● Configure I2C bus parameters, including clock speed and addressing modes

Qualification Pack

PC9. ● Develop and debug code to implement I2C communication between devices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Understand the fundamental concepts of UART communication, including start/stop bits, parity, and baud rate.
 - Know the various configuration settings of UART such as baud rate, parity, stop bits, and data bits
- KU2.**
- Understand the principles of SPI communication, including the roles of master and slave devices, and data exchange mechanisms.
 - Learn the configuration settings of SPI, such as clock polarity, clock phase, and data order

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Expertise in connecting and configuring communication interfaces such as UART, SPI, and I2C with microcontrollers and peripherals.
 - Deep understanding of various communication protocols and their practical implementation
- GS2.**
- Ability to plan and manage embedded systems projects, including setting timelines and milestones.
 - Skill in executing projects efficiently, ensuring all communication components are correctly integrated and functioning
- GS3.**
- Strong ability to work effectively within a team, collaborating with peers, and sharing knowledge.
 - Clear communication skills to convey complex technical information to team members and stakeholders

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>UART (Universal Asynchronous Receiver-Transmitter) Communication</i>	30	25	-	6
PC1. • Demonstrate understanding of UART communication principles, including start/stop bits and data frames	-	-	-	-
PC2. • Utilize software tools to configure UART settings on various microcontrollers	-	-	-	-
PC3. • Write and debug code to implement UART communication between a microcontroller and peripheral devices	-	-	-	-
<i>SPI (Serial Peripheral Interface) Communication</i>	30	25	-	7
PC4. • Explain the principles and components of SPI communication, including master/slave roles and data exchange mechanisms	-	-	-	-
PC5. • Write and test code to enable SPI communication between microcontrollers and peripherals	-	-	-	-
PC6. • Develop a project showcasing effective SPI communication in an embedded system application	-	-	-	-
<i>I2C (Inter-Integrated Circuit) Communication</i>	40	30	-	7
PC7. ● Describe the fundamentals of I2C communication, including addressing and data transfer mechanisms	-	-	-	-
PC8. ● Configure I2C bus parameters, including clock speed and addressing modes	-	-	-	-
PC9. ● Develop and debug code to implement I2C communication between devices	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0211
NOS Name	Embedded Communication Protocols (UART, SPI, I2C, CAN, USB)
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0212: Wireless Sensor networks

Description

This NOS provides a comprehensive introduction to Wireless Sensor Networks (WSN), covering fundamental concepts, architectures, communication technologies, protocols, and advanced topics. Designed for students and professionals, the course aims to equip participants with the knowledge and skills required to design, develop, and implement efficient and secure WSN applications across various domains.

Scope

The scope covers the following :

- The scope of this course is to provide learners with a comprehensive understanding of the fundamental concepts, architectures, and technologies underpinning WSNs. It emphasizes hands-on learning through programming exercises and practical case studies to illustrate real-world applications. The curriculum covers essential topics such as WSN architecture, communication technologies, protocol stacks, security challenges, energy efficiency, and advanced topics like QoS and cross-layer optimization.

Elements and Performance Criteria

WSN Protocol Stack

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate a comprehensive understanding of the functions and interactions between the Physical, MAC, Network, Transport, and Application layers.
- PC2.** Accurately differentiate between CSMA, TDMA, and Hybrid protocols, explaining their respective advantages and limitations in various WSN scenarios.
- PC3.** Develop and simulate routing protocols (Proactive, Reactive, and Hybrid) for different network configurations, demonstrating an understanding of their operational principles

WSN Security and Energy Efficiency

To be competent, the user/individual on the job must be able to:

- PC4.** Identify and articulate the primary security challenges in WSNs, including potential vulnerabilities and attack vectors
- PC5.** Implement encryption, authentication, and key management techniques in WSN scenarios to ensure data integrity and security
- PC6.** Apply optimization techniques and energy-aware protocols to design and manage WSNs that minimize power consumption while maintaining performance

WSN Operating Systems and Programming

To be competent, the user/individual on the job must be able to:

- PC7.** Explain the key features and functionalities of various WSN operating systems, such as Contiki OS and TinyOS
- PC8.** Utilize appropriate platforms, languages, and tools to write and debug WSN applications, demonstrating practical programming skills

Qualification Pack

- PC9.** Complete hands-on exercises that involve setting up and programming WSN nodes, showcasing the ability to apply theoretical knowledge in real-world scenarios

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Develop a thorough understanding of the protocol stack for wireless sensor networks, including the functions and interactions of the Physical, MAC, Network, Transport, and Application layers
- KU2.** Acquire knowledge about the primary security challenges in wireless sensor networks, including common vulnerabilities and types of attacks
- KU3.** Gain an understanding of the features and functionalities of WSN operating systems such as Contiki OS and TinyOS, including their architectures and use cases

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Enhance skills in troubleshooting network issues, diagnosing protocol errors, and optimizing network performance
- GS2.** Improve skills in documenting system architectures, protocol configurations, and programming code to ensure clarity and reproducibility
- GS3.** Foster an independent learning attitude to explore new tools, operating systems, and programming languages relevant to WSN

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>WSN Protocol Stack</i>	30	25	-	6
PC1. Demonstrate a comprehensive understanding of the functions and interactions between the Physical, MAC, Network, Transport, and Application layers.	-	-	-	-
PC2. Accurately differentiate between CSMA, TDMA, and Hybrid protocols, explaining their respective advantages and limitations in various WSN scenarios.	-	-	-	-
PC3. Develop and simulate routing protocols (Proactive, Reactive, and Hybrid) for different network configurations, demonstrating an understanding of their operational principles	-	-	-	-
<i>WSN Security and Energy Efficiency</i>	30	25	-	7
PC4. Identify and articulate the primary security challenges in WSNs, including potential vulnerabilities and attack vectors	-	-	-	-
PC5. Implement encryption, authentication, and key management techniques in WSN scenarios to ensure data integrity and security	-	-	-	-
PC6. Apply optimization techniques and energy-aware protocols to design and manage WSNs that minimize power consumption while maintaining performance	-	-	-	-
<i>WSN Operating Systems and Programming</i>	40	30	-	7
PC7. Explain the key features and functionalities of various WSN operating systems, such as Contiki OS and TinyOS	-	-	-	-
PC8. Utilize appropriate platforms, languages, and tools to write and debug WSN applications, demonstrating practical programming skills	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Complete hands-on exercises that involve setting up and programming WSN nodes, showcasing the ability to apply theoretical knowledge in real-world scenarios	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0212
NOS Name	Wireless Sensor networks
Sector	Electronics
Sub-Sector	
Occupation	IoT Solution
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0213: Linux Device Driver Development

Description

This NOS offers a comprehensive introduction to Linux Kernel architecture and device driver development. Designed for software engineers, system administrators, and Linux enthusiasts, the course covers essential concepts, practical implementations, and debugging techniques related to device drivers in the Linux environment

Scope

The scope covers the following :

- The scope of this course is to equip learners with the essential knowledge and skills required to understand, develop, and troubleshoot device drivers in the Linux operating system

Elements and Performance Criteria

Kernel Module Basics:

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate the ability to compile, load, and unload kernel modules successfully in a Linux environment
- PC2.** Modify and test kernel module parameters to observe changes in kernel behavior and functionality
- PC3.** Identify common errors during module loading and unloading, and troubleshoot issues effectively using debugging tools

Character Device Drivers

To be competent, the user/individual on the job must be able to:

- PC4.** Develop character device drivers that support open, close, read, and write operations, ensuring proper functionality and error handling
- PC5.** Utilize the ioctl system call effectively within character device drivers to manage device-specific functionalities and configurations
- PC6.** Allocate and manage major and minor numbers for character devices accurately, ensuring proper device registration and accessibility

Block Device Drivers

To be competent, the user/individual on the job must be able to:

- PC7.** Implement I/O operations (request and response handling) in block device drivers, ensuring efficient data transfer and error handling
- PC8.** Optimize buffer cache usage and I/O scheduling algorithms in block device drivers to enhance data transfer performance and system responsiveness
- PC9.** Analyze the performance of block device drivers under different load conditions, identify bottlenecks, and propose optimization strategies for improved performance

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understanding of the process of compiling, loading, and unloading kernel modules
- KU2.** Understanding of character devices and their interaction with user-space applications
- KU3.** Understanding of data structures used for block devices and their functions in handling I/O operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to identify and troubleshoot issues related to kernel module loading, character device operations, and block device performance
- GS2.** Clear and effective communication of technical concepts related to kernel modules, device drivers, and system performance
- GS3.** Evaluating different approaches to kernel module customization, device operation implementation, and performance optimization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Kernel Module Basics:</i>	30	25	-	6
PC1. Demonstrate the ability to compile, load, and unload kernel modules successfully in a Linux environment	-	-	-	-
PC2. Modify and test kernel module parameters to observe changes in kernel behavior and functionality	-	-	-	-
PC3. Identify common errors during module loading and unloading, and troubleshoot issues effectively using debugging tools	-	-	-	-
<i>Character Device Drivers</i>	30	25	-	7
PC4. Develop character device drivers that support open, close, read, and write operations, ensuring proper functionality and error handling	-	-	-	-
PC5. Utilize the ioctl system call effectively within character device drivers to manage device-specific functionalities and configurations	-	-	-	-
PC6. Allocate and manage major and minor numbers for character devices accurately, ensuring proper device registration and accessibility	-	-	-	-
<i>Block Device Drivers</i>	40	30	-	7
PC7. Implement I/O operations (request and response handling) in block device drivers, ensuring efficient data transfer and error handling	-	-	-	-
PC8. Optimize buffer cache usage and I/O scheduling algorithms in block device drivers to enhance data transfer performance and system responsiveness	-	-	-	-
PC9. Analyze the performance of block device drivers under different load conditions, identify bottlenecks, and propose optimization strategies for improved performance	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0213
NOS Name	Linux Device Driver Development
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0214: Internet of Things

Description

This NOS provides a comprehensive understanding of the Internet of Things (IoT) landscape. The course covers the evolution, applications, challenges, and opportunities of IoT. It delves into IoT architecture, communication models, protocols, devices, sensors, and security measures. Students will engage in hands-on exercises with IoT platforms, middleware, and data management tools, exploring real-world case studies and developing practical IoT applications.

Scope

The scope covers the following :

- The scope of this course is to prepare students for various roles in the IoT industry, including IoT development, system integration, data analysis, and project management, fostering a deep understanding of the interconnected world of IoT

Elements and Performance Criteria

IoT Architecture and Communication Models

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and explain various real-world applications of IoT across different industries
- PC2.** Describe the architecture of an IoT system and the role of each component
- PC3.** Implement basic IoT communication using one of the studied protocols in a hands-on exercise

IoT Devices and Sensors

To be competent, the user/individual on the job must be able to:

- PC4.** Demonstrate the setup and configuration of IoT devices in a practical project
- PC5.** Design a small-scale IoT project incorporating appropriate sensors based on project requirements
- PC6.** Implement a simple edge computing solution as part of a hands-on project

IoT Security and Privacy

To be competent, the user/individual on the job must be able to:

- PC7.** Identifying IoT Security Threats and Vulnerabilities and list common security threats and vulnerabilities in IoT systems
- PC8.** Describe advanced security measures such as blockchain, secure boot, and secure firmware updates
- PC9.** Apply advanced security techniques to an IoT project to enhance its security posture.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Gain a detailed understanding of the components of an IoT system, including sensors, actuators, gateways, and their interactions

Qualification Pack

- KU2.** Understand the principles of operation, characteristics, and common types of sensors used in IoT (e.g., environmental, biomedical, motion sensors).
- KU3.** Understand common security threats and vulnerabilities associated with IoT systems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to analyze complex problems, identify key issues, and devise effective solutions in the context of IoT
- GS2.** Improve the ability to communicate technical information effectively to both technical and non-technical stakeholders, fostering collaboration and teamwork
- GS3.** Learn to allocate and manage resources effectively, including hardware, software, and human resources, to ensure project success

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>IoT Architecture and Communication Models</i>	30	25	-	6
PC1. Identify and explain various real-world applications of IoT across different industries	-	-	-	-
PC2. Describe the architecture of an IoT system and the role of each component	-	-	-	-
PC3. Implement basic IoT communication using one of the studied protocols in a hands-on exercise	-	-	-	-
<i>IoT Devices and Sensors</i>	30	25	-	7
PC4. Demonstrate the setup and configuration of IoT devices in a practical project	-	-	-	-
PC5. Design a small-scale IoT project incorporating appropriate sensors based on project requirements	-	-	-	-
PC6. Implement a simple edge computing solution as part of a hands-on project	-	-	-	-
<i>IoT Security and Privacy</i>	40	30	-	7
PC7. Identifying IoT Security Threats and Vulnerabilities and list common security threats and vulnerabilities in IoT systems	-	-	-	-
PC8. Describe advanced security measures such as blockchain, secure boot, and secure firmware updates	-	-	-	-
PC9. Apply advanced security techniques to an IoT project to enhance its security posture.	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0214
NOS Name	Internet of Things
Sector	Electronics
Sub-Sector	
Occupation	IoT Solution, Industrial IoT (IIoT)
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0215: AI for Embedded Application

Description

This NOS provides a comprehensive introduction to Artificial Intelligence (AI) with a focus on its applications in embedded systems. Starting with an overview of AI and its various applications, students will delve into the basics of machine learning (ML) and deep learning (DL), understanding neural networks and their components. The course then addresses the unique challenges of running AI models on embedded devices, covering hardware considerations for AI acceleration in embedded systems

Scope

The scope covers the following :

- The course aims to cover the basics of machine learning and deep learning, introduce neural networks and their components, and delve into the challenges of running AI models on embedded devices.

Elements and Performance Criteria

Introduction to Artificial Intelligence

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to explain the history and evolution of AI, including its key milestones and applications
- PC2.** Demonstrate understanding of AI challenges such as ethical considerations, bias, and data privacy
- PC3.** Analyze case studies showcasing AI applications in various fields like healthcare, finance, and autonomous systems

Embedded AI Frameworks and Tools:

To be competent, the user/individual on the job must be able to:

- PC4.** Implement AI models using TensorFlow Lite, PyTorch, or ONNX on embedded devices
- PC5.** Evaluate the performance of AI frameworks on hardware accelerators like Edge TPU or NVIDIA Jetson
- PC6.** Compare and contrast different frameworks and tools based on their suitability for specific embedded AI applications

Integration of AI with Embedded Systems

To be competent, the user/individual on the job must be able to:

- PC7.** Successfully integrate sensors with embedded systems and collect sensor data for AI model input
- PC8.** Develop and deploy AI-enabled features on embedded platforms, ensuring real-time data processing and low latency
- PC9.** Optimize AI models for resource-constrained environments such as microcontrollers, focusing on efficiency and accuracy

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understanding the core concepts of artificial intelligence, machine learning, and deep learning
- KU2.** Knowledge of embedded systems architecture, including processors, memory, and I/O interfaces.
- KU3.** Understanding of ONNX format for representing deep learning models and its interoperability with different frameworks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze and solve complex problems related to AI model deployment on embedded devices
- GS2.** Effective communication skills for presenting AI concepts and solutions to technical and non-technical stakeholders
- GS3.** Flexibility to adapt to evolving AI technologies, frameworks, and hardware platforms in the embedded space

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Artificial Intelligence</i>	30	25	-	6
PC1. Ability to explain the history and evolution of AI, including its key milestones and applications	-	-	-	-
PC2. Demonstrate understanding of AI challenges such as ethical considerations, bias, and data privacy	-	-	-	-
PC3. Analyze case studies showcasing AI applications in various fields like healthcare, finance, and autonomous systems	-	-	-	-
<i>Embedded AI Frameworks and Tools:</i>	30	25	-	7
PC4. Implement AI models using TensorFlow Lite, PyTorch, or ONNX on embedded devices	-	-	-	-
PC5. Evaluate the performance of AI frameworks on hardware accelerators like Edge TPU or NVIDIA Jetson	-	-	-	-
PC6. Compare and contrast different frameworks and tools based on their suitability for specific embedded AI applications	-	-	-	-
<i>Integration of AI with Embedded Systems</i>	40	30	-	7
PC7. Successfully integrate sensors with embedded systems and collect sensor data for AI model input	-	-	-	-
PC8. Develop and deploy AI-enabled features on embedded platforms, ensuring real-time data processing and low latency	-	-	-	-
PC9. Optimize AI models for resource-constrained environments such as microcontrollers, focusing on efficiency and accuracy	-	-	-	-
NOS Total	100	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0215
NOS Name	AI for Embedded Application
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0219: Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver

Description

This integrated project involves developing a real-time embedded system using ARM Cortex-M series microcontrollers for sensor data acquisition. The system will leverage Linux Internals and Real-Time Operating Systems (RTOS) for efficient task management and performance. Communication protocols including UART, SPI, I2C, CAN, and USB will be utilized for seamless data exchange. Wireless Sensor Networks will be incorporated to enable remote monitoring and data collection. Additionally, custom device drivers will be implemented to interface with various sensors and peripherals, ensuring robust and scalable functionality.

Scope

The scope covers the following :

- The project scope includes designing and implementing a real-time embedded system using ARM Cortex-M microcontrollers, integrating advanced sensor data acquisition with Linux Internals and RTOS for optimized performance. It will incorporate communication protocols such as UART, SPI, I2C, CAN, and USB to ensure efficient data transfer. The system will also integrate Wireless Sensor Networks for remote monitoring and data collection. Custom device drivers will be developed to interface with sensors and peripherals, enhancing the system's functionality and scalability.

Elements and Performance Criteria

Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver

To be competent, the user/individual on the job must be able to:

- | | |
|-------------|--|
| PC1. | Ability to demonstrate precise data acquisition from various sensors with minimal error and reliable output. |
| PC2. | Successful integration of sensor data into the embedded system, showing seamless interaction between sensors and the ARM Cortex-M microcontroller. |
| PC3. | Proficiency in implementing and configuring communication protocols (UART, SPI, I2C, CAN, USB) for effective data transfer between components |
| PC4. | Skill in developing custom device drivers that ensure proper functionality and compatibility with various sensors and peripherals |
| PC5. | Effectiveness in using RTOS for task scheduling, ensuring that real-time operations are performed accurately and efficiently. |
| PC6. | Demonstrated ability to integrate RTOS functionalities with Linux systems, showcasing understanding of both environments and their interaction. |

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understanding different types of sensors (e.g., temperature, humidity, pressure) and their functions, including how to select and interface them with microcontrollers.
- KU2.** Knowledge of communication protocols (UART, SPI, I2C, CAN, USB) including their operational principles, data transfer mechanisms, and application in embedded systems.
- KU3.** Understanding how to integrate RTOS functionalities with Linux systems, including managing resources and ensuring seamless operation across both environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Skill in addressing challenges related to sensor interfacing and data processing, ensuring reliable system performance.
- GS2.** Flexibility in adapting to different protocols and hardware requirements, and applying best practices for development and integration
- GS3.** Efficiently managing time to handle real-time tasks and system operations, ensuring timely execution and resource allocation.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver</i>	-	-	80	20
PC1. Ability to demonstrate precise data acquisition from various sensors with minimal error and reliable output.	-	-	-	-
PC2. Successful integration of sensor data into the embedded system, showing seamless interaction between sensors and the ARM Cortex-M microcontroller.	-	-	-	-
PC3. Proficiency in implementing and configuring communication protocols (UART, SPI, I2C, CAN, USB) for effective data transfer between components	-	-	-	-
PC4. Skill in developing custom device drivers that ensure proper functionality and compatibility with various sensors and peripherals	-	-	-	-
PC5. Effectiveness in using RTOS for task scheduling, ensuring that real-time operations are performed accurately and efficiently.	-	-	-	-
PC6. Demonstrated ability to integrate RTOS functionalities with Linux systems, showcasing understanding of both environments and their interaction.	-	-	-	-
NOS Total	-	-	80	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0219
NOS Name	Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver
Sector	Electronics
Sub-Sector	
Occupation	Embedded System Design
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

Qualification Pack

- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns the roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds

Qualification Pack

and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to the syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0206.Embedded System Design & C-Programming	100	80	0	20	200	9
NIE/ELE/N0207.Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming	100	80	0	20	200	9

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0208.Sensor data acquisition and actuation in embedded systems	100	80	0	20	200	9
NIE/ELE/N0209.Linux Internals	100	80	0	20	200	9
NIE/ELE/N0210.RTOS for Embedded Application	100	80	0	20	200	9
NIE/ELE/N0211.Embedded Communication Protocols (UART, SPI, I2C, CAN, USB)	100	80	0	20	200	9
NIE/ELE/N0212.Wireless Sensor networks	100	80	0	20	200	9
NIE/ELE/N0213.Linux Device Driver Development	100	80	0	20	200	9
NIE/ELE/N0214.Internet of Things	100	80	0	20	200	9
NIE/ELE/N0215.AI for Embedded Application	100	80	0	20	200	9
NIE/ELE/N0219.Implementation of project based on Core ARM Cortex M series, Sensor data acquisition, Linux Internals, RTOS, Communication Protocols (UART, SPI, I2C, CAN, USB), Wireless Sensor Networks, and Device Driver	0	0	80	20	100	8
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	2
Total	1020	830	80	220	2150	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf